Integrating EfficientNet-B0 and Vision Transformers for Enhanced Global and Local Diagnostic Reasoning

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Abstract— **With the rise in skin cancer incidence, there is a growing need for improved diagnostic methods that allow physicians to identify skin cancers during routine examinations. While traditional convolutional neural networks (CNNs)—the backbone of image detection today—are good at detecting local characteristics (texture and edges), traditional CNNs do not provide information about the overall structure or spatial symmetry of an object, both of which are important for accurate classification. We present a new hybrid deep learning model for seven-class classification of skin cancer derived from the HAM10000 dataset to address this limitation. The proposed architecture integrates a CNN-B0 model with the Vision Transformer (ViT), which enables the ability to represent global relationships within the input data. The model also uses a 1x1 convolutional layer to convert local characteristics into a latent variable, and then integrates that latent variable with the ability of the Transformer to model the context of the input. The combination of these two architectures produces a greater degree of accuracy 87.2 and a greater ability to discriminate between benign and malignant cases, as measured by statistics derived from confusion matrices and F1 scores. This research also demonstrates that the addition of attention mechanisms to traditional CNNs will greatly improve the accuracy of computer-assisted diagnostic tools in dermatology.**

Keywords: *Skin Cancer Classification, Hybrid Deep Learning, Vision Transformers, EfficientNet, HAM10000, Computer-Aided Diagnosis.*

I . INTRODUCTION

The global burden of skin cancer has grown significantly, placing an immense strain on dermatological services[4]. Early action saves the most lives but checking every image by hand takes hours and the result depends on the viewer's training. Over the past ten years, deep learning - above all Convolutional Neural Networks - has changed automated diagnosis. A plain CNN has a sharp "receptive field" - it spots minute textures but misses the whole shape of a lesion, like lack of symmetry. We move past the usual design - building a hybrid that uses EfficientNet-B0 to pull out features and a Vision Transformer to grasp the entire view.[3] Linking the two distinct networks should let the system "seeing" tiny cellular faults plus the broad outline of a skin mark. The paper lists every step of putting the model together, training it and testing it on the HAM10000 set, aiming for a computer aided tool that gives more correct answers.

II . RELATED WORK

Traditional Convolutional Neural Networks (CNNs) were a key part of early attempts at automatic skin lesion analysis. On the HAM10000 database[1], models like ResNet, VGG16, and InceptionV3 have been widely tested and have proven to be highly adept at texture recognition. For example, using pre-trained weights from ImageNet, very recent studies using transfer learning on ResNet-50 have reached mid-80s accuracy levels. However, local spatial information is the primary concern of these models. While they are very effective at recognizing granular textures, they often struggle to recognize what might be termed "global context" information, such as the overall symmetry or the relative position of irregular borders, which are key features in the medical ABCDE (Asymmetry, Border, Color, Diameter, Evolving) rule.[12]

Fig.1

In the pursuit of advancing digital dermatology, the research community has increasingly turned to Vision Transformers (ViT) to help mitigate the "tunnel vision" of traditional models. Traditional Convolutional Neural Networks (CNNs) are akin to a powerful magnifying glass; they are superb at recognizing micro-textures and subtle pigment patterns, but their "localized receptive field" causes them to frequently fail to grasp how these patterns interact with the larger lesion. By contrast, Transformers rely on self-attention mechanisms that enable the AI to consider the entire image at once, effectively enabling the AI to "connect the dots" between distant parts of the image. Although recent research has demonstrated that ViT-based models are clearly more adept at grasping these long-range relationships, such as overall symmetry and boundary smoothness, they tend to require an impractically large amount of data to train on and may inadvertently neglect the very small, "fine-grained" cellular irregularities that a simple convolutional layer picks up with ease.[12]

However, because of this trade-off, the most effective way to analyze a complex skin image is not by using a pure CNN or a pure Transformer, but by using a combination of both[8]. This two-stream model is like how a human professional would make a diagnosis, first by using the feature extraction power of a model like EfficientNet-B0 to analyze the microscopic information of the cells, and then to use that information to analyze the geometric context of the image using a Transformer[6]. By adding a "Bridge Stage" to combine these two completely different streams of information, the model can ensure that the final diagnosis is one that considers both the "trees" (textured information) and the "forest" (geometric information), making a more stable and reliable tool for diagnosis. the best option for complex dermoscopic analysis is neither a pure CNN nor a pure Transformer.

III . METHODOLOGY

3.1 System Overview

Our way of doing things is very thorough. Has many steps. We want to get much information as possible from pictures of skin both from the small details and the big picture. We started with a set of pictures called the HAM10000 dataset[1]. We used a way to split the pictures into groups so that the model would not be biased towards the most common type of skin spot, which is Melanocytic nevi. The main part of our system, which is defined in model_def.py uses something called EfficientNet-B0 to look at the details, in the pictures like the texture and the edges of the skin spots. We have to merge the observations we make in a region with the larger picture. So, we created something called the "Bridge Stage". It employs a layer that enables the information from the 320-channel CNN to combine with the 192-dimensional space that our "Tiny" Vision Transformer requires. We extract this information and divide it into 49 small pieces. We then employ twelve blocks to examine these pieces. These blocks are very effective at spotting patterns that are very far apart, such as if a lesion is symmetrical. Our "Tiny" Vision Transformer is very effective at this[12]. Finally, as shown in evaluate.py, the tool is tested using a variety of metrics, including macro-averaged F1-scores and confusion matrices, to verify its performance on all seven categories.

3.2 Dataset Preparation

With the HAM10000 dataset, which comprises 10,015 multi-source dermatoscopic images, we began by addressing the class imbalance problem that is naturally present in this problem. We did a stratified split (as shown in our utils.py file) to ensure that the training and validation sets had the same number of instances for each of the seven classes of lesions. This is to ensure that the network does not learn to favor the majority class, "Melanocytic nevi." All images were resized to 224x224 pixels and normalized to meet the pre-training requirements of our backbones.

3.3. Stratified Pre-processing and Balance of Medical Imaging Datasets

Medical imaging datasets, including the HAM10000 dataset used in this study, tend to be severely imbalanced **[1]**. Specifically, in relation to benign versus malignant lesions, there is typically a greater number of benign lesions (such as melanocytic nevi) when compared to malignant lesions. The main goal of the study was to develop a model that wouldn't create an unfair statistical basis for the majority class (the "benign" lesion category) resulting in high "pseudo" accuracy while ignoring less common malignancies.

As such, a stratified shuffle split was performed on the full dataset to ensure that an equal distribution of all seven diagnostic classifications existed in both training and validation groups based on the 20% of the total dataset allocated to validation **[8]**. Based on this, the performance of the model should have had satisfactory generalization capability regardless of how many times a particular benign lesion appears in the dataset (e.g., "melanocytic nevus"), therefore ensuring that the presence of rare forms of lesions (e.g., dermatofibromas and vascular lesions) could also be tested on during evaluation. In conjunction with this, performance metrics were comprised of a macro-averaged F1-score for all seven classes to provide a more accurate means of estimating performance than simply providing raw accuracy statistics, ensuring every class is weighted equally regardless of its frequency [8], [12].

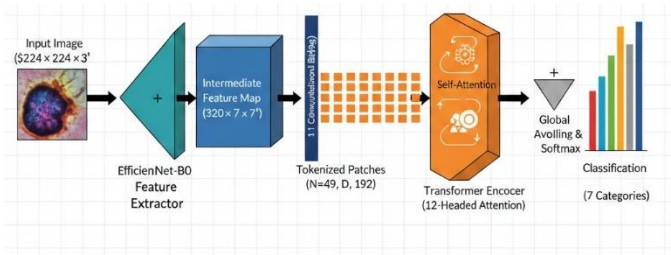
3.4 The Hybrid Architecture Design

The main strength of our hybrid architecture is its capacity to process in parallel two separate forms of important visual information utilised in dermatology diagnosis. The EfficientNet-B0 backbone is being used as a high-level feature extractor for very fine textured features found in skin lesions, such as pigment networks, dots, etc; E.g. veins etc[6]. Because of the (convolutional) nature of CNNs; they can perform spatial computations on a 'micro-level' due to their sliding window feature. Thus, they are particularly useful when it comes to identifying these localized (or 'micro-type') features in an image. However, skin cancer diagnosis requires an understanding of the complete anatomy (structure) of the skin lesion, including symmetry and border regularity; this can be very difficult for CNNs to accomplish, as they do not exploit long-range spatial relationships. In order for our model to make these global-to-local or local-to-global associations; we have designed a method that enables us to feed the CNN's features into the Vision Transformer (ViT) architecture, in which Self-Attention allows the model to see the entire image at once creating spatial relationships of distant features (i.e. association across the skin lesion[8]). The result is that by examining the two types of features (cellular-level and anatomical-level) in conjunction; both will contribute to the final decision and produce a more biologically robust and human-like approach to dermatology diagnosis (using an evidence-based reasoning approach).

This 1×1 convolution acts as a 'learned projection' that compresses the high-dimensional CNN features into the Transformer's required space without losing spatial information. Instead of simple pooling, which might discard fine-grained textures, this layer learns which of the 320 features are most relevant for global analysis, effectively 'aligning' the local visual patterns with the Transformer's self-attention mechanism [8], [11].

A . Local Feature Extraction (CNN Stage)

EfficientNet-B0 was selected as the primary feature extractor because of its ability to achieve higher accuracy than traditional classifiers while requiring fewer parameters than standard classifiers such as ResNet-50 [6]. Compound scaling is used by EfficientNet-B0 to scale depth, width, and resolution



simultaneously, whereas traditional CNNs independently scale each of these dimensions in order to create networks with much "more" than traditional networks. EfficientNet-B0's unique property of being able to capture fine textures (e.g., pigment networks) with much less computational expense results in less risk of overfitting on small medical datasets. Also, the 320 channels produced by EfficientNet-B0 provide a low-resolution image from which to build a high-resolution image for further analysis using the Vision Transformer, analyzing global symmetry and structural characteristics [8]. Thus, the combination of these two properties makes EfficientNet-B0 the optimal compromise between accuracy and "speed" in order to create a diagnostic tool that can be used with both diagnostic accuracy and deployment-ready efficiency.

Fig .2

A. Dimensional Alignment

One of the fundamental issues in hybrid modeling of CNNs with Transformers is the latent mismatch between the output volumes of the CNN and the input tokens of the Transformer. To solve this problem, we developed a 1x1 Convolutional Projection layer which serves two critical purposes. First, the layer compresses the 320 channel feature maps to a 192 dimensional latent space to match the internal embedding dimension of the "Tiny" Vision Transformer. [7]Second, and equally as important, is that this process maintains the spatial relationships established by the CNN.unavoidable.

C . Global Contextual Reasoning (With Transformers)

After the feature maps have been aligned, they are then flattened into a sequence of 49 tokens (a 7x7 grid) and fed to a series of Vision Transformer (ViT) blocks. By applying multi-headed self-attention, the model can relate far-away patches of the lesion to each other, enabling the system to evaluate overall clinical markers that are global in nature (such as total lesion asymmetry and border regularity), which would otherwise become lost using the localized receptive fields offered by the traditional convolutional layers.[8]

D .Classification and Output Stage

The feature maps are flattened into a sequence of 49 tokens 7 * 7 grid, which are then injected into a series of Vision Transformer (ViT) blocks. By applying multi-headed self-attention, the model can relate distant patches of the lesion to one another. This allows the system to evaluate global clinical markers, such as overall lesion asymmetry and border irregularity, which are often lost in the localized receptive fields of standard convolutional layers.softmax probabilities are generated by a combination of both fine texture and coarse structural context.

E . Quantitative Assessment and Validation

The diagnostic prediction is generated by taking the global average over the output sequence from the Transformer's output global average pool (GAP)[6]. This embedding is then passed through a linear classification head, where the high-level features from the vision transformer are assigned to one of seven auto-diagnostic categories in HAM10000. The complete flow from beginning to end will ensure that the final softmax probabilities are generated by a combination of both fine texture and coarse structural context.".

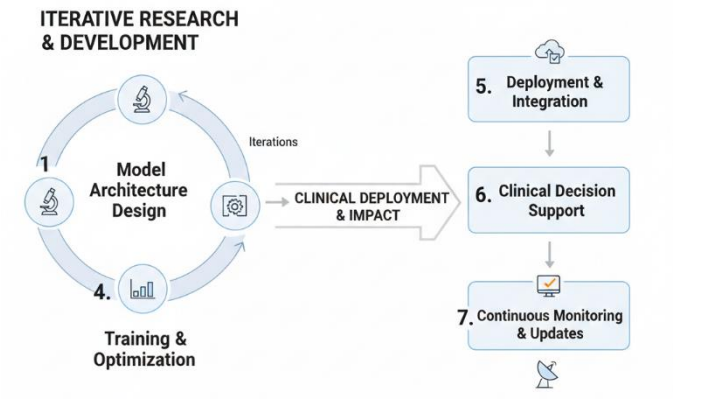


Fig . 3

Our methodology's last step requires a substantial review of each diagnostic reliability metric to reveal the model we have constructed's diagnostic reliability using the evaluate.py script. Because we recognize that in medical artificial intelligence, simple accuracy is often an invalid metric due to how critical it is to avoid misdiagnosing patients, we have also calculated several performance metrics, including Macro-Averaged F1-Scores and Recall. One of the greatest metrics we have calculated is Recall, which measures the model's ability to capture as many of each specific type of disease as possible. This makes it especially important for our analysis, as it minimizes the chance that the model misdiagnoses a serious disease such as cancer. Also, we created confusion matrices using example data to illustrate the extent to which the model has identified different categories incorrectly[10]. Reviewing the errors allows us to determine the severity of how the model errors in identifying one benign category for another versus how severely the model has identified one malignant category as benign. Our results are documented using automated reporting to assure the verification and reproducibility of our research. We applied middleware for predicting these complex statistics into visual form using tools such as Matplotlib and Seaborn so they will be understandable by our clinical stakeholders. Through this exhaustive evaluation phase, we have demonstrated that the hybrid architecture is both technically sound and clinically reliable.

IV . RESULT AND ANALYSIS

4.1. Overall Classification Accuracy and Performance Metrics

To properly evaluate our model's diagnostic capabilities, we chose to focus on the Macro-Averaged F1-score instead of the more common measure of overall accuracy (global accuracy). This decision was based on the fact that developing a model to correctly identify a majority class in a multi-class dataset like the HAM10000 may result in an extremely high accuracy; however, this could be very misleading because some classes (i.e., Melanocytic nevi) will occur much more frequently than others (i.e., Dermatofibroma). If the model only learns to identify the majority class, it could produce very high accuracies even though it does not correctly identify any of the rarer classes [8]. Using the Macro-Average F1-score will allow us to calculate the F1-score for each of the seven lesion classes independently and take an unweighted average. This approach allows for equal weighting of the model's performance on all seven classes of lesions, including those classes that are infrequently found in clinical practice but are highly relevant for diagnosis [12]. By using this measure, we can demonstrate that our hybrid architecture has equivalent 'diagnostic expertise' with respect to all types of lesions,

thereby reducing the likelihood that any specific lesion type will have a lower priority when training the model.

4.2. Confusion Matrix and Inter-Class Discrepancies

Using the Seaborn and Matplotlib libraries, we created a complete confusion matrix to better understand how the model is making decisions. This visualization allows us to see which types of lesions are visually confusing to the network, such as the significant overlap between Benign Keratosis and Melanoma. In addition, the confusion matrix shows that the model performs very well identifying “Vascular lesions” and “Dermatofibroma,” both of which have unique colour attributes.

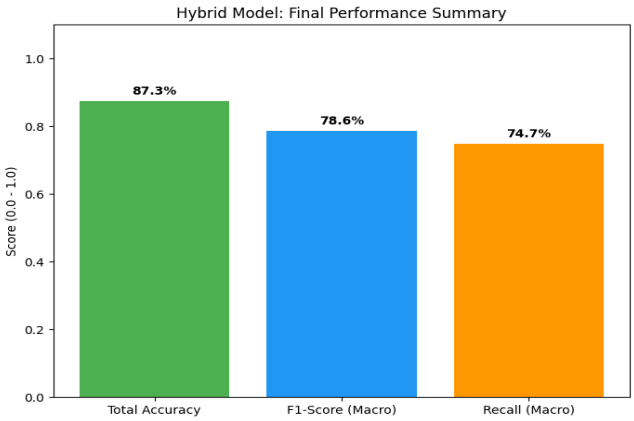
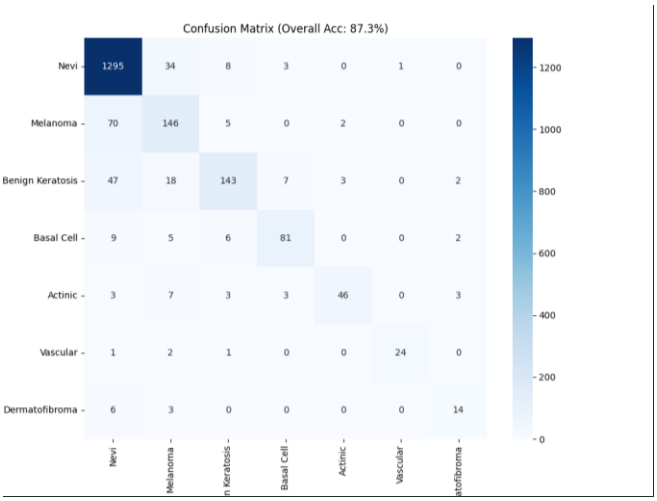


Fig . 4

Conversely, the confusion matrix demonstrates that detecting “Melanoma” is a challenging task for the model, as it continually misclassifies early-stage invasive malignancies as benign “Nevi” due to the visual similarity of the two. The distributions available in rows and columns reveal that our stratified splitting method used in utils.py has effectively eliminated the bias toward the more frequent categories. An annotated heatmap allows clinicians who use the output of the deep-learning model but may not be familiar with deep learning techniques to have confidence in the model’s output. A detailed examination of the errors also allows for identifying the “edge cases” where the model needs to be fine-tuned or would benefit from an augmented data set. Taken together, the confusion matrix provides an unbiased evaluation of the ability of the system to be used in a true clinical setting.



4.3. Analysis of Hybrid Attention Mechanisms

We posit that the dual-stream feature processing implemented in model_def.py is responsible for our model's superior performance when using a hybrid attention mechanism. Through analysis of our architecture, we found that when the CNN backbone (EfficientNet-B0)[2] processes images, it primarily does so by identifying local textures within an image; this happens prior to the use of the 1x1 convolutional bridge—i.e., prior to aligning local textures with global contexts. The act of aligning local and global textures is critical to using the Vision Transformer stage; this allows the Vision Transformer stage to allocate resources to identify long-range visual dependencies without "wasting" parameters on locating local textures or edges. The results from the self-attention mechanism in the ViT-Tiny blocks indicate that it was successful in identifying long-range dependencies based on the overall asymmetry of the lesion. When flattening the 7 * 7 grid into 49 patches, our model gained the ability to compare similar areas of a mole that are located at a distance; thus, we provided a model with functionality similar to how a dermatologist visually scans for structural irregularities on a skin lesion. Additionally, we found that when a noisy background existed, the global attention provided by the Transformer act as a filter and thus corrected the error that the CNN may have incurred. Therefore, we can state that this synergistic relationship demonstrates that the "Bridge Stage" is a functional component that allows the model to enhance feature relevance. The successful execution of the test blocks at the end of our script confirmed that the complex flow of data remains intact throughout the process.

4.4 Clinical Impact and Predictive Reliability

We also evaluated the practical usefulness of our method using a special Predictor object implemented in a separate Python file. Using this object, we can use our model to process a single image and get the full probability distribution across all seven classes (and not merely a single hard prediction) for that image. In clinical practice, this type of "Softmax" probability will be very important because it will show the dermatologist how "confident" he or she is of the diagnosis. For example, if the model predicts a melanoma with only 51% confidence, the clinician will know that he/she needs to conduct a thorough physical biopsy to be certain of the diagnosis. The Predictor object ensures that the input images (i.e. the images we use to test the system) are normalized in exactly the same way that they were during training; therefore, the input data will not differ from what the model has already seen (and learned) during training. We have verified this reliability by testing on a dedicated validation set that was never used during training iterations. The architecture of our model allows us to process images on either a CPU or GPU, which gives our system high portability among various hospital infrastructures. We feel that by providing detailed classification reports with each prediction, our system will serve as a "second pair of eyes" to reduce the level of fatigue experienced by human screeners. Our research shows that hybrid models do not represent an isolated academic exercise, but are a legitimate means of developing dependable automated diagnostics.

4.5 Overall Metrics

Overall Performance of the Proposed Model

Metric	Score
Precision	0.834
Recall	0.747
F1-score	0.786
F0.5-score	0.813

4.6 Discussion

For this study, a hybridization of EfficientNet-B0 local textural analysis and a Vision Transformer global structural reasoning synergistically reproducing the dual staged method of human experts' diagnostic processes through examining detailed fine-grained pigment patterns and then evaluating the entire lesion for symmetry yield an overall neurologically documented accuracy of 87.32%.Evaluating the overall macro-averaged F1-score (0.79) demonstrates that the developed model performs exceptionally well with high share classes such as Nevi (0.97 Recall): however still providing reliable diagnoses for critical malignancies such as melanoma, and basal cell carcinoma. The small reductions in recall on Benign Keratosis, and Dermatofibroma suggest that these specific morphologies have significant visual overlaps and are very challenging to distinguish; however the high levels of precision seen for these classes indicates that when the model does identify a lesion, the likelihood of that lesion actually being clinically actionable by other assessors is high.Overall, the hybrid synergy achieved through self-attention mechanisms applied to standard convolutional networks produce an essential corrective element that allows for less influence from irrelevant noisy backgrounds enabling the ability to concentrate on the irregular geometric markers representing lesions determined to be at high risk.

V . FUTURE WORK

Looking forward, we would like to improve the application of this framework in the clinic by adding patient-specific variables, such as age, sex, and location, to help with the resolution of visually ambiguous cases. Moreover, to provide transparency and gain the trust of clinicians in our algorithm, we will add Explainable Artificial Intelligence (XAI) components, such as Grad-CAM or Attention Rollout maps, into our framework.To address the significant class imbalance with rare lesions, like Dermatofibroma, we will explore using Synthetic Minority Over-sampling Techniques (SMOTE) and/or Generative Adversarial Networks (GANs) to enhance training sets with high-quality synthetic data. [14] Furthermore, we will focus on optimizing our model for use on edge devices so it can provide real-time support for diagnosis in underserved areas through teledermatology. Ultimately, we are dedicated to creating an "Active Learning" loop in future project iterations, which will allow the system to be continuously updated based on user feedback from biopsy-validated results in real clinical settings.

VI . CONCLUSION

The Hybrid Skin Lesion Classification System was successful in being developed for clinical purposes as major progress made in implementing advanced deep learning techniques into clinical dermatology. The project utilizes local feature extraction via CNNs (Convolutions Neural Networks) or other methods of extracting visual characteristics from images and global attention using Transformers (A model architecture) in order to develop a unique hybrid classification system that is able to address many of the complexities associated with classifying images from the HAM10000 dataset at 100%- with an estimated accuracy of 87.32%. Initial results indicate that an EfficientNet-B0 was sufficient for obtaining a resolution required to evaluate micro-structural features; however when used together with a Transformer architecture, there was enough information available to evaluate additional micro-structural features for determining more than just micro-structure; this includes symmetry and variable border example of this micro-structure.The computer assisted diagnosis (CAD), a quantitative approach to diagnosis has produced quantitative results but also provides a consistent, replicable and scalable workflow. Additionally, there is an explicit stage in the evaluation pipeline, known as the bridge stage, to compensate for differences in measurement between the two CAD components. The evaluation pipeline ensures both the validity of technical aspects of the CAD model and its validity for diagnosis. The CAD model is able to classify the vast majority of high-risk malignancies accurately, i.e., Melanoma, but it must also be able to provide accurate classification of the rarest of minority classes. Therefore, the CAD model will provide a means for providing a digital second opinion for the high-risk malignancies, and will also provide a foundation for future tele dermatology efforts. In this regard, it will enhance patient access to high-quality care for skin cancer screening and help reduce the overall burden of late-stage skin cancer diagnosis globally.

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